

HDV PLATOONING

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WHAT IS HDV PLATOONING?



HDV platooning is the linking of two or more trucks in convoy, using connectivity technology and automated driving support systems.

These vehicles automatically maintain a set, close distance between each other when they are connected for certain parts of a journey, for instance on highways.

The truck at the head of the platoon acts as the leader, with the vehicles behind reacting and adapting to changes in its movement – requiring little to no action from drivers.

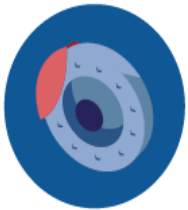
WHAT ARE THE BENEFITS OF PLATOONING?

CLEAN



- Platooning lowers fuel consumption. Given that trucks can drive closer together, the air-drag friction is reduced significantly.
- Platooning can reduce CO2 emissions by up to 16% from the trailing vehicles and by up to 8% from the lead vehicle (according to the recent ITS4CV study by Ertico).

SAFE



- Platooning helps to improve safety. Accelerating and Braking are automatic and immediate; the trucks following the lead vehicle only need one-fifth of the time a human would need to react.

EFFICIENT



- Platooning optimizes transport by using roads more effectively, or increasing the road capacity, delivering goods faster and reducing traffic jams.

STATEMENTS

► Objectives

- The purpose of this project is to understand the modeling and control theory of HDV platooning by reviewing achievements in research area.
- As a background research to support Applied Project.

► Constraints

- Modeling and simulations are not included.
- Not for real-world experiments or validations.

► Assumptions

- All Vehicles are in same specs and remain in platooning all the time.
 - Lateral control and longitudinal control will be studied independently.
 - Vehicles out of platooning will not be taken in to consideration.
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HOW HDV PLATOONING IS ACHIEVED AND IMPROVED?

► Longitudinal Control

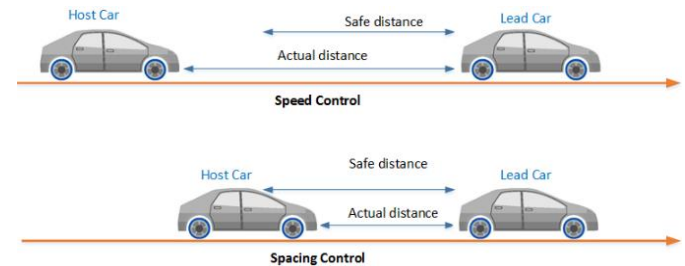
- Constant inter-truck distance
- Constant-time headway
(Distance on the basis of velocity)

► Lateral Control

- Model Predictive Control (MPC)

► Optimal Platooning

- Road Topography Preview
- Look-Ahead Control (LAC)



LONGITUDINAL CONTROL

► Truck model

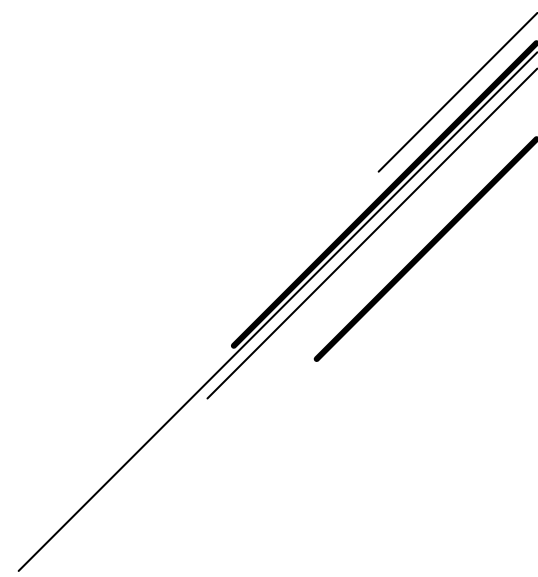
$$\begin{aligned}\dot{x}_i &= v_i, \\ \dot{v}_i &= -\frac{1}{T_i}v_i + \frac{K_i}{T_i}u_i\end{aligned}$$

- If $u_i \geq 0$, u_i is throttle position, T_i is T_{ai} , and K_i is K_{ai} . The input is $100u_i(\%)$. If $u_i < 0$, u_i is pressure on the brake, T_i is T_{bi} , and K_i is K_{bi} . The input in the case is $|u_i|$ (MPa).

► Longitudinal control

- Error system in distance and velocity

$$\begin{aligned}e_i &= x_i - x_{i+1} - d - L_i, \\ w_i &= v_i - v_r,\end{aligned}$$



LONGITUDINAL CONTROL

► Front model controller

$$u_1 = \frac{T_i}{K_i} \left(\frac{1}{T_i} v_i + \dot{v}_r - c_0 w_1 \right)$$

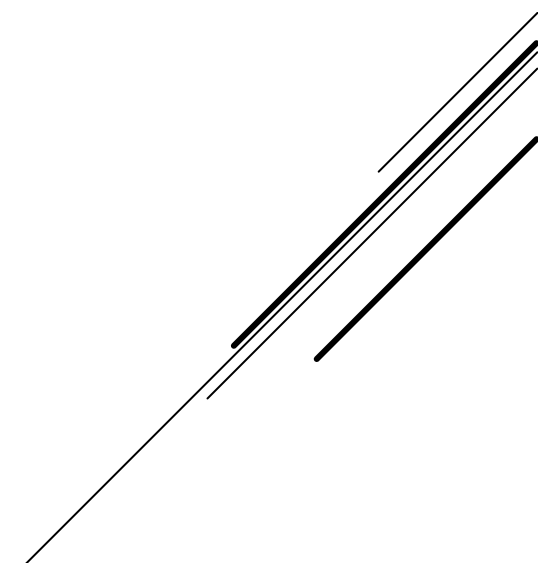
$$u_i = \frac{T_i}{K_i} \left(\frac{1}{T_i} v_i + \dot{v}_r - c_0 w_i + k_1 e_{i-1} + c_1 (v_{i-1} - v_i) \right)$$

► Front-Rear model controller

$$u_1 = \frac{T_i}{K_i} \left(\frac{1}{T_i} v_i + \dot{v}_r - c_0 w_1 - k_1 e_i - c_1 (v_1 - v_2) \right)$$

$$u_i = \frac{T_i}{K_i} \left(\frac{1}{T_i} v_i + \dot{v}_r - c_0 w_i + k_1 e_{i-1} - k_2 e_i \right. \\ \left. + c_1 (v_{i-1} - v_i) - c_1 (v_i - v_{i+1}) \right),$$

$$u_n = \frac{T_i}{K_i} \left(\frac{1}{T_i} v_i + \dot{v}_r - c_0 w_n + k_1 e_{n-1} + c_1 (v_{n-1} - v_n) \right)$$



LONGITUDINAL CONTROL

► Front-Rear model controller stability

- Lyapunov function candidate V

$$V = \frac{k_1}{2} \sum_{i=1}^{n-1} e_i^2 + \frac{1}{2} \sum_{i=1}^n w_i^2 \quad \dot{V} = -c_0 \sum_{i=1}^n w_i^2 - c_1 \sum_{i=1}^{n-1} (v_i - v_{i-1})^2 \leq 0$$

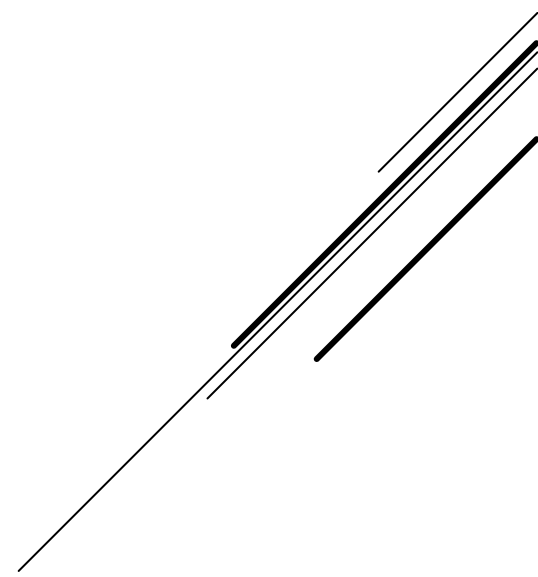
- w_i converges to 0 and d_i is bounded.

$$\dot{w}_1 = -c_0 w_1 - k_1 e_1 - c_1 (w_1 - w_2)$$

$$\dot{w}_j = -c_0 w_j + k_1 e_{j-1} - k_1 e_j + c_1 (w_{j-1} - w_j) - c_1 (w_j - w_{j+1}), 2 \leq j \leq n-1,$$

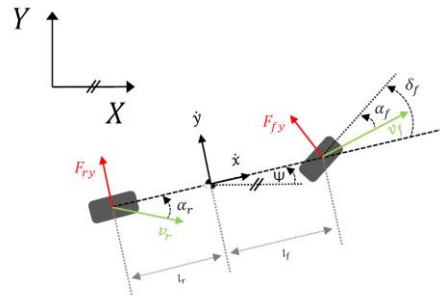
$$\dot{w}_n = -c_0 w_n - k_1 w_n e_{n-1} + c_1 (w_{n-1} - w_n).$$

- e_{n-1} sequentially converge to 0 sequentially because w_i and $\dot{w}_i$ converge to 0.



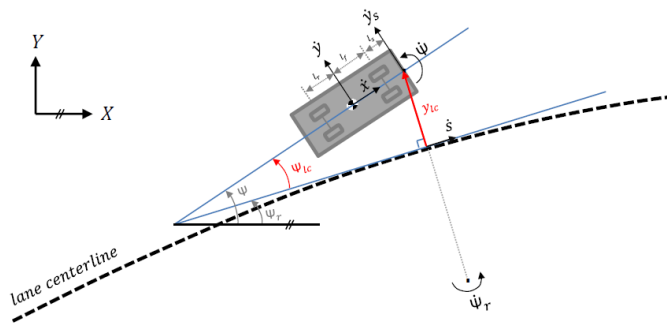
LATERAL CONTROL

► Lateral vehicle dynamics



$$\begin{bmatrix} \ddot{y} \\ \ddot{\Psi} \end{bmatrix} = \begin{bmatrix} \frac{C_f + C_r}{m\dot{x}} & -\frac{C_f l_f - C_r l_r}{m\dot{x}} - \dot{x} \\ -\frac{(C_f l_f - C_r l_r)}{I_z \dot{x}} & -\frac{C_f l_f^2 + C_r l_r^2}{I_z \dot{x}} \end{bmatrix} \begin{bmatrix} \dot{y} \\ \dot{\Psi} \end{bmatrix} + \begin{bmatrix} \frac{C_f}{m} \\ \frac{C_f l_f}{I_z} \end{bmatrix} \delta_f$$

► Lateral position dynamics



$$\begin{bmatrix} \ddot{y} \\ \ddot{\Psi} \\ \dot{\Psi}_{cl} \\ \dot{y}_{cl} \end{bmatrix} = \begin{bmatrix} \frac{C_f + C_r}{m\dot{x}} & -\frac{C_f l_f - C_r l_r}{m\dot{x}} - \dot{x} & 0 & 0 \\ -\frac{(C_f l_f - C_r l_r)}{I_z \dot{x}} & -\frac{C_f l_f^2 + C_r l_r^2}{I_z \dot{x}} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & l_f + l_s & \dot{x} & 0 \end{bmatrix} \begin{bmatrix} \dot{y} \\ \dot{\Psi} \\ \Psi_{lc} \\ y_{lc} \end{bmatrix} + \begin{bmatrix} \frac{C_f}{m} \\ \frac{C_f l_f}{I_z} \\ 0 \\ 0 \end{bmatrix} \delta_f + \begin{bmatrix} 0 \\ 0 \\ \dot{x} \\ 0 \end{bmatrix} \kappa$$

The curvature κ is the feed forward input disturbance.

LATERAL CONTROL

► Model Predictive Control (MPC)

- Discretization

$$\dot{x}(t) = A_c x(t) + B_c u(t) \quad \longrightarrow \quad e^{-A_c t} \dot{x}(t) - e^{-A_c t} A_c x(t) = e^{-A_c t} B_c u(t)$$

- Prediction model

$$z(k+1) = A_d z(k) + B_d u(k) \quad A_d = e^{A_c T} \approx \sum_{i=0}^{i_{\text{sufficient}}} \frac{(A_c T)^i}{i!} \quad B_d = \int_0^T e^{A_c \Upsilon} B_c d\Upsilon \approx \sum_{i=0}^{i_{\text{sufficient}}} \frac{(A_c T)^i}{(i+1)!} B_c T$$

- Constrained quadratic cost function

$$J(k) = \sum_{i=0}^{N_h-1} \|z(k+1+i) - z_{\text{ref}}(k+1+i)\|_Q^2 + \sum_{i=0}^{N_u-1} \|u(k+i) - u_{\text{ref}}(k+i)\|_R^2$$

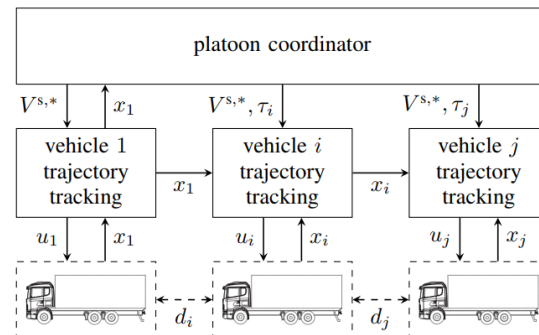
- Control inputs

$$u^*(k) = \min_u J(k)$$

► Kalman Filter may be added as to process noise.

OPTIMAL PLATOONING

- ▶ Road Topography Preview – uphill, downhill, etc.
- ▶ Look-Ahead Control (LAC)
 - A Look-Ahead controller uses knowledge about the HDVs position and future road topography when controlling the vehicles driving strategy.
 - An optimal velocity trajectory is determined for the vehicle and its given route in order to lower fuel consumption.



[4] Optimal Platooning of Heavy-Duty Vehicles, Rikard Ohlsén and Erik Sten, Department of Electrical Engineering, Linköping University, 2018

[5] Fuel-efficient heavy-duty vehicle platooning by look-ahead control, Valerio Turri, Bart Besselink, Jonas Martensson, Karl H. Johansson

QUESTIONS?

